



4116 - Close up samples of Exoplanetary Systems: Characterizing the next Interstellar Object

Cycle: 2, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Karen J. Meech (PI)	University of Hawaii
Dr. Jacqueline Keane (CoI) (Contact)	University of Hawaii
Dr. Jan Kleyna (CoI)	University of Hawaii
Dr. Robert Weryk (CoI) (CSA Member)	The University of Western Ontario
Dr. Richard Wainscoat (CoI)	University of Hawaii
Olivier Raoul Hainaut (CoI) (ESA Member)	European Southern Observatory - Germany
Dr. Marco Micheli (CoI) (ESA Member)	ESA SSA-NEO Coordination Centre
Dr. Davide Farnocchia (CoI)	Jet Propulsion Laboratory
Prof. Alan Fitzsimmons (CoI) (ESA Member)	The Queen's University of Belfast
Dr. Matthew M Knight (CoI)	United States Naval Academy
Dr. Dennis Bodewits (CoI)	Auburn University
Dr. Sean Raymond (CoI) (ESA Member)	Universite de Bordeaux
Dr. Marc W. Buie (CoI)	Southwest Research Institute
Darryl Seligman (CoI)	Cornell University
Dr. Bin Yang (CoI)	Planetary Science Institute

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
ISO				
	2	NIRSpec ISO	NIRSpec IFU Spectroscopy	(1) ISO-23.5MAG

Folder	Observation	Label	Observing Template	Science Target
	6	NIRSpec ISO-Background	NIRSpec IFU Spectroscopy	(1) ISO-23.5MAG
	3	NIRCam ISO	NIRCam Imaging	(1) ISO-23.5MAG

ABSTRACT

One of the most significant recent findings in astronomy has been the discovery of two interstellar objects (ISOs) passing through our solar system. The two ISOs discovered to date are strikingly different: 2I/Borisov appeared cometary, while 1I/ʻOumuamua looked asteroidal with characteristics that still are not fully explained, but could have been with the JWST program proposed here.

A detailed look at ISOs will provide unparalleled nearby access to the chemical and physical conditions of exoplanet formation. Comets and some asteroids are the largely unaltered remnants of the planetary accretion process, tracing both dust and volatiles - H₂O, CO, CO₂ - in the disk. JWST will reveal detailed information about ISOs that we cannot obtain from the ground, such as size and albedo, properties of solid ices and surface materials, and simultaneous measurements of water, CO and CO₂.

Accordingly, with this 17.4 hr program, we request one ToO (>14 d) to characterize an ISO discovered in Cycle 2. The observing program will depend on whether the next ISO is like 2I/Borisov or 1I/ʻOumuamua. IR observations will allow us to estimate an albedo and size. NIRSpec spectra will detect H₂O, CO, CO₂, and water ice bands, while NIRCam deep imaging will search for faint coma

OBSERVING DESCRIPTION

We have designed a non-disruptive ToO program, with a response time set to >14 days, the minimum value for a non-disruptive ToO.

The NRSIRS2RAPID readout mode was chosen for the NIRSpec PRISM observations to maximize the SNR in a short amount of time. For the medium resolution filters, NRSRAPID readout mode is chosen to maximize the S/N and to stay within the Data Volume Limit. To mitigate the effects of detector cosmetics we will use a 4-point nodding pattern. In the APT, we have implemented the WATA target acquisition mode and used the NIRSpec target_acq template in the ETC to confirm that the signal-to-noise is sufficient for WATA. However, if the GTO and ERS programs show that chance acquisition of background objects is a problem then we will use NONE as the target acquisition mode.

Deep NIRCam imaging with the F210M filter will be used to search for faint coma. We will use 2 sub-pixel dithers and the MEDIUM8 readout pattern

Special Requirements: This is an On Hold non-disruptive ToO program with a response time $\geq$ 14 days, with the sequence of observations set as non-interruptible.

Proposal 4116 - Targets - Close up samples of Exoplanetary Systems: Characterizing the next Interstellar Object

Generic Targets	#	Name	Criteria	Description
	(1)	ISO-23.5MAG	Interstellar Object	
	Comments: Non-sidereal object.			

Proposal 4116 - Observation 2 - Close up samples of Exoplanetary Systems: Characterizing the next Interstellar Object

Observation	Proposal 4116, Observation 2: NIRSpec ISO										Tue May 30 17:03:52 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over lower threshold											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria									Description
	(1)	ISO-23.5MAG	Interstellar Object									
	Comments: Non-sidereal object.											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	F140X	NRSRAPID	3	1	1	42.947	61999	
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	NRSRAPID	120	1	false	true	NONE	4	4	5196.597	
	2	G395M/F290LP	NRSRAPID	120	2	false	true	NONE	4	8	10393.193	
	3	PRISM/CLEAR	NRSRAPID	60	1	false	true	NONE	4	4	2619.772	
Special Requirements	On Hold ToO											
	Target Of Opportunity Response Time 14 Days, Number of Activations 1											
	Sequence Observations 2, 3, 6, Non-interruptible											

Proposal 4116 - Observation 6 - Close up samples of Exoplanetary Systems: Characterizing the next Interstellar Object

Observation	Proposal 4116, Observation 6: NIRSpec ISO-Background										Tue May 30 17:03:52 GMT 2023	
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy											
	(Visit 6:1) Warning (Form): Data Excess over lower threshold											
Generic Targets	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	ISO-23.5MAG	Interstellar Object									
Acquisition	Comments: Non-sidereal object.											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	F140X	NRSRAPID	3	1	1	42.947	61999	
Dithers	#	Dither Type		Size	Starting Point			Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	NRSRAPID	120	1	false	true	NONE	4	4	5196.597	
	2	G395M/F290LP	NRSRAPID	120	2	false	true	NONE	4	8	10393.193	
	3	PRISM/CLEAR	NRSRAPID	60	1	false	true	NONE	4	4	2619.772	
Special Requirements	On Hold ToO											
	Sequence Observations 2, 3, 6, Non-interruptible											

Proposal 4116 - Observation 3 - Close up samples of Exoplanetary Systems: Characterizing the next Interstellar Object

Observation	Proposal 4116, Observation 3: NIRCam ISO Diagnostic Status: Warning Observing Template: NIRCam Imaging										Tue May 30 17:03:52 GMT 2023
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria	Description							
	(1)	ISO-23.5MAG	Interstellar Object	Comments: Non-sidereal object.							
Template	Module	Subarray			Target Placement						
	B	FULL			Module Gap						
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F356W	MEDIUM8	8	3	6	2	5067.755		
Special Requirements	On Hold ToO Sequence Observations 2, 3, 6, Non-interruptible										